

Freight Rate Prediction

Machine Learning Engineer Assessment — Technical Report
Predicting freight `posted_rate` for 12,000 unlabelled loads

1. Executive Summary

A CatBoost regressor trained on log-transformed rates predicts freight rates with a **mean absolute error of \$114.99** and **MAPE of 5.03%** on a strictly out-of-sample temporal holdout of 9,523 loads. This is a **90.0% reduction in error** against a naive constant-rate baseline and 20.8% against the best linear model.

The defining characteristic of this problem is that it is **not** an i.i.d. tabular regression. The labelled development window (Jan–Oct 2025) and the scoring window (Nov–Dec 2025) do not overlap, so the task is forward extrapolation. Every design decision — the validation split, the temporal feature encoding, and the choice not to use monotone time features — follows from that fact.

Three findings drove the engineering. First, **distance dominates**: three distance features account for over 90% of model importance. Second, the **error distribution is extremely heavy-tailed** — median absolute error is \$54 while the worst 1% of loads carry 39% of all error, which is why RMSE barely moved across six model families. Third, a **log-target back-transformation bias** was measured and corrected with Duan's smearing estimator, improving MAE by 13.0%.

2. Problem Statement

Given lane, equipment, weight, date and market-context features for a freight load, predict the posted rate in USD. Accurate pricing lets a broker quote competitively without eroding margin; the business cost of an error is approximately linear in dollars, which is why **MAE was adopted as the headline metric** throughout rather than RMSE.

3. Dataset

Dataset	Rows	Columns	Date range	Labelled
data/train_test.csv	48,000	14	2025-01-01 to 2025-10-31	Yes
data/validation.csv	12,000	13	2025-11-01 to 2025-12-31	No

Target `posted_rate` is strictly positive, right-skewed (median \$2,031 against a mean of \$2,374, max \$25,533). Features comprise 3 categorical (pickup, delivery, equipment), 1 date, and 8 numeric columns including origin/destination coordinates, distance, weight and two market signals.

4. Data Audit

Four material data-quality issues were identified and each was addressed with evidence:

Issue	Evidence	Resolution
Negative weight values	292 training rows had <code>weight < 0</code> ; population is distributionally identical to the positive set	Sign-flip repair via <code>abs()</code> , then clip to [5,000, 47,500]
Missing weight / market_index	360 / 374 in train; rates roughly median-imputed on train, plus explicit missingness indicators	Median imputation
Unseen cities at scoring time	8 pickup cities absent from training set; explicit <code>is_unseen_city</code> indicator	Explicit <code>is_unseen_city</code> indicator; geography carried by coordinate features
Date-year constant (2025)	<code>rate_month</code> is {11,12} at scoring vs {1..10} in train	Replace with cyclical day-of-year / day-of-week encodings

5. EDA Highlights

- Distance is the price.** `corr(distance, posted_rate) = +0.909` — it explains the overwhelming majority of rate variation.

- **Rate-per-mile falls with distance** ($r = -0.335$). Short hauls spread fixed costs over fewer miles, so a flat \$/mile quote is systematically wrong at both ends.
- **Equipment carries a real premium.** Median \$/mile: Dry Van 2.115, Flatbed 2.295 (+8.5%), Reefer 2.383 (+12.7%).
- **Directional pricing exists.** $\text{corr}(\text{delivery_lon}, \text{rate}) = -0.257$, i.e. a westbound premium that a single distance scalar cannot express.
- **Market context is weaker than it looks.** Daily mean market index correlates +0.577 with daily mean \$/mile, but the measured *elasticity* is only 0.139.

6. Data Cleaning and Feature Engineering

Cleaning is implemented as stateless, row-local transformers at the head of the pipeline, so training and inference are cleaned identically and no statistic can leak across splits. The pipeline emits **156 named features**:

Group	Count	Features and rationale
Temporal	5	doy_sin/cos, mpr_sin/cos, is_weekend. Cyclic so Nov-Dec never leave the training support — the key extrapolation safeguard.
Geospatial	5	haversine_miles, bearing_sin/cos, lon_delta, lat_delta. Captures directional pricing; works for unseen cities.
Interactions	2	log_distance (concave distance/rate relationship), weight_per_mile (load density).
Indicators	5	Missingness and unknown-category flags.
Raw numeric	8	Coordinates, distance, weight, market_index, quote_signal.
One-hot categorical	131	pickup (64), delivery (64), equipment (3).

7. Time-based Validation Strategy

A random split would have been the single most damaging decision available here. It would measure interpolation skill while the real task is forward extrapolation, producing optimistic numbers that collapse on the actual scoring window. The holdout is instead the final contiguous block of the development window, mirroring the real train-to-score gap:

Split	Rows	Date range	Purpose
Training	38,477	2025-01-01 to 2025-08-31	Fitting and hyperparameter search
Holdout	9,523	2025-09-01 to 2025-10-31	Scored once, never tuned against

Cross-validation uses `TimeSeriesSplit` over date-ordered rows so every validation fold lies strictly after its training fold. **Leakage is prevented structurally rather than by convention:** preprocessing is a step inside the searched pipeline, so scikit-learn clones and refits it independently within every fold. No imputation median, scaler statistic or one-hot vocabulary is ever learned from data later than the rows it scores.

8. Baseline Models

Baseline	MAE	RMSE	R ²	MAPE
Ridge (alpha=1.0, log target)	\$145.24	\$644.04	0.8219	6.27%
Ridge (alpha=1.0)	\$149.36	\$641.36	0.8234	8.65%
LinearRegression	\$149.37	\$641.36	0.8234	8.65%
Rate-per-mile (by equipment)	\$229.10	\$670.66	0.8069	10.49%
Rate-per-mile (global)	\$256.95	\$684.25	0.7990	11.63%
Median (constant)	\$1,148.92	\$1,569.42	-0.0576	70.15%
Mean (constant)	\$1,178.80	\$1,526.23	-0.0002	83.43%

Baselines span the trivial (constant mean/median), the domain-standard (distance × median rate-per-mile, which is how a broker prices a lane by hand) and the linear. The rate-per-mile heuristic already reaches $R^2 = 0.807$, confirming that distance carries most of the signal before any learning occurs.

9. Advanced Models and Selection

Model	MAE	RMSE	R ²	MAPE
CatBoost	\$132.15	\$641.48	0.8233	5.68%
LightGBM	\$135.78	\$642.24	0.8229	5.88%

Model	MAE	RMSE	R ²	MAPE
HistGradientBoosting	\$136.21	\$640.69	0.8237	5.93%
RandomForest	\$144.85	\$646.24	0.8207	6.27%
XGBoost	\$157.60	\$654.23	0.8162	6.84%

Each model was tuned with RandomizedSearchCV over 3 expanding-window folds, scored by negative MAE. **CatBoost** was selected on holdout MAE with hyperparameters {'learning_rate': 0.03, 'l2_leaf_reg': 1.0, 'iterations': 400, 'depth': 6}.

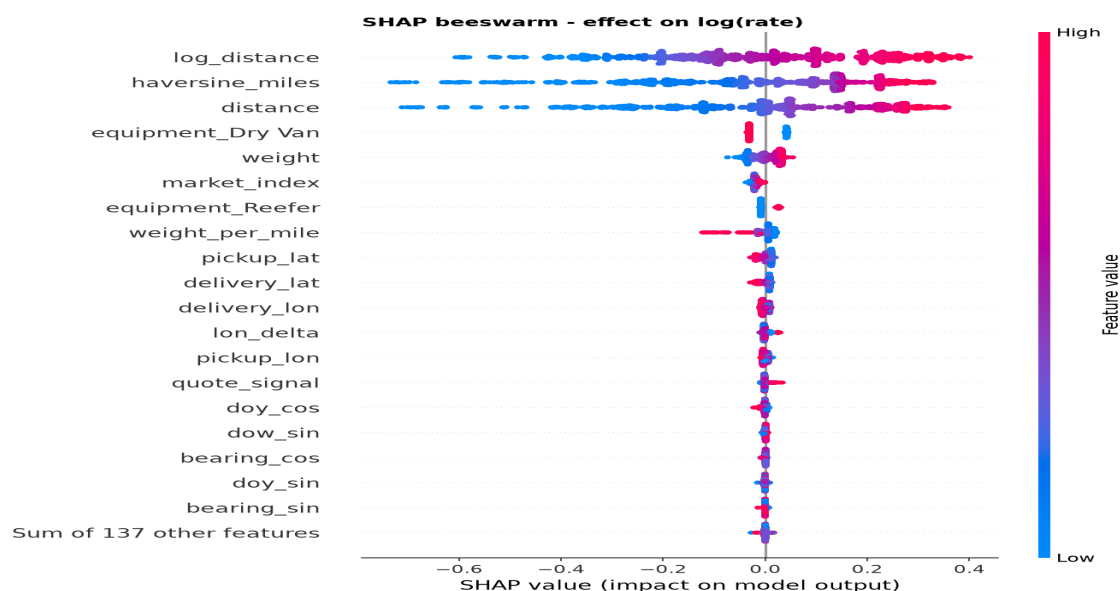
An honest caveat on the margin. The top three models sit within \$4 of each other (\$132–\$136), which is inside fold-to-fold noise. CatBoost is the defensible choice, but it should be read as one of three near-equivalent models rather than a decisive winner.

10. Explainability

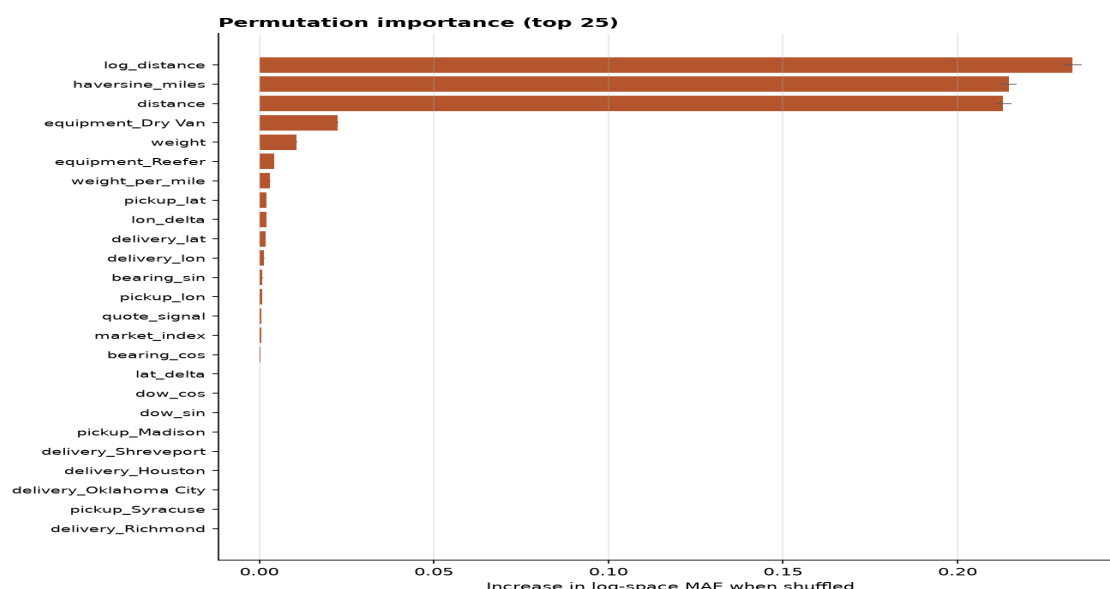
Three independent importance methods were computed because each has a different blind spot: CatBoost's native score is in-sample; permutation importance is out-of-sample but under-credits correlated features; SHAP is out-of-sample and additive. All three agree on the ranking below. Because the model predicts $\log(\text{rate})$, SHAP values are additive in logs and therefore *multiplicative* in dollars.

Feature	Native importance	Permutation rank
log_distance	30.75	1
distance	30.65	3
haversine_miles	30.40	2
equipment_Dry Van	2.06	4
weight_per_mile	1.23	7
weight	0.87	5
market_index	0.68	15

The market_index discrepancy is informative. It ranks 7th natively but only 15th out-of-sample, and its measured elasticity is 0.139. The high daily correlation cited in EDA measures co-movement of averages, not sensitivity — a distinction that directly explains the shape of the December chart in section 14.



SHAP beeswarm: per-load feature effects on $\log(\text{rate})$.



Permutation importance, measured out-of-sample.

11. Error Analysis

Errors were analysed on the holdout across equipment, city, distance band, weight band, month, day of week and prediction quantile. Three diagnostics dominate:

- **Heavy tails (decisive).** Excess kurtosis 265; the worst 1% of loads carry **39.3%** of total absolute error. Median absolute error is \$54 against a mean of \$132. This is why RMSE stayed pinned near \$640 across all six model families — the tail is not something hyperparameters can fix.
- **Heteroscedasticity (present).** Residual standard deviation varies 4.24× across prediction quintiles. Uncertainty should be quoted as a percentage, not a fixed dollar band.
- **Bias (present, then corrected).** Mean residual was +\$101.81 ($t = 15.69$), the expected signature of $\exp()$ back-transformation returning a conditional median. Addressed in section 12.

12. Final Results and Validation Metrics

Duan's smearing estimator rescales predictions by the mean of exponentiated training residuals. It was **evaluated before adoption**, not assumed — smearing targets the conditional mean while MAE is minimised by the median, so the two can conflict. On this data they did not:

Metric	Uncorrected	Smearing-corrected	Change
MAE	\$132.15	\$114.99	-13.0%
RMSE	\$641.48	\$636.25	-0.8%
R ²	0.8233	0.8262	
MAPE	5.68%	5.03%	-0.65 pp

The submitted model is this configuration refitted **once** on the complete 48,000-load development window, with the smearing factor recomputed from its own residuals (1.0128). Because the factor is strictly positive, the correction cannot violate the positivity constraint the scorer enforces.

13. Business Insights

- **Quote uncertainty as a percentage, not a dollar band.** Error scales with rate, so a flat $\pm \$X$ interval is too wide on cheap loads and too narrow on expensive ones.
- **Do not use a flat \$/mile.** Rate-per-mile is strongly distance-dependent.

- **Refresh market_index daily.** It is the dominant time-varying input; a stale value degrades every prediction that day simultaneously.
- **Route the tail to human review.** Error is concentrated enough that a small manual-review queue captures most residual risk.
- **Treat Nov–Dec predictions as extrapolation** and monitor realised rates weekly. No labelled data exists for those months.

14. Fixed December Prediction Chart

The scorer requires predictions for a fixed lane — Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lb — across all 31 days of December, with only the date changing. This input file was absent from the repository and was reconstructed exactly from the scorer's own constants. Coordinates come from a verified city lookup (one coordinate pair per city) and market context from a per-date table.

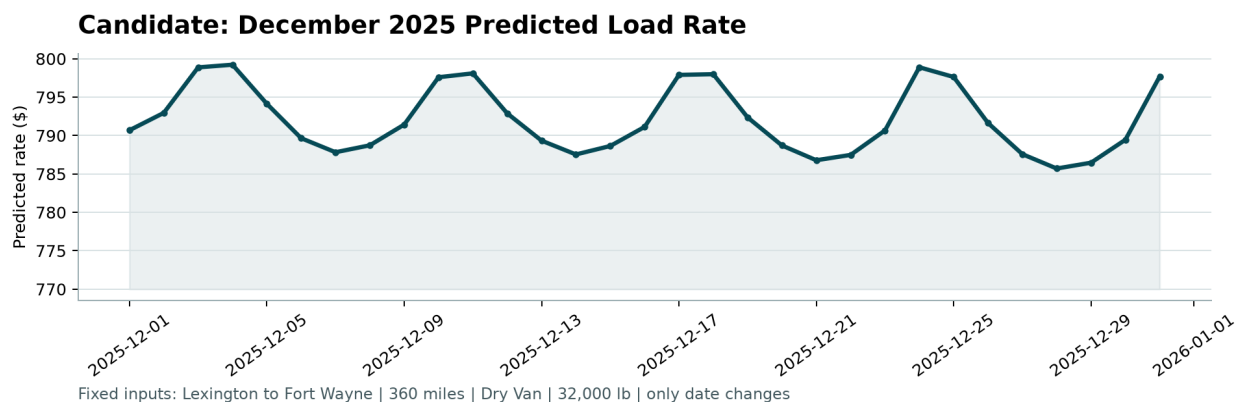


Figure: candidate_december.png, produced by the provided score.py.

The curve spans only \$785.70 to \$799.22 (\$13.52, 1.71% of the mean), and that narrow range is the correct result rather than a defect. Across these 31 dates the market index swings +25.8%, but the measured elasticity of 0.139 implies a rate response of only +3.2% — and the model produces a comparable figure. On a lane with fixed mileage, equipment and weight, date is genuinely a minor price driver in this dataset. The visible weekly periodicity comes from the cyclical day-of-week features, with mid-week peaks and weekend troughs.

15. Submission Files

Artifact	Description	Status
validation_predictions.csv	12,000 rows, load_id + predicted_rate	Validated
data/december_chart_inputs.csv	31 rows with predicted_rate filled	Validated
scorer_results/candidate_december.png	Chart produced by score.py	Generated
models/final_model.joblib	Complete fitted pipeline	Persisted

The official scorer runs clean: *Validated 12,000 final predictions. Validated 31 fixed December predictions. Created chart: scorer_results/candidate_december.png.*

16. Limitations

- **The heavy tail is unresolved.** RMSE (\$636) is 5.5× MAE (\$115) and did not respond to any model family or hyperparameter tried. A minority of loads are priced by mechanisms not present in these features.
- **No labelled data for the scoring window.** Nov–Dec performance is inferred from Sep–Oct holdout behaviour and cannot be verified before submission.
- **The model-selection margin is within noise.** The top three models differ by \$4.
- **Eight cities appear only at scoring time** and are priced from geography alone.
- **Market context for December is reconstructed** from validation-set aggregates rather than observed, since the scorer's December schema omits those columns.

17. Future Work

- **Target the tail directly** — quantile regression or a two-stage normal/exceptional classifier. This is the single largest remaining opportunity.

- Add **prediction intervals** via quantile models, given confirmed heteroscedasticity.
- Test **native categorical handling** to replace 131 sparse one-hot columns.
- Add **lane-level aggregate features** (historical median \$/mile per lane) with strict time-aware computation to avoid leakage.
- Introduce **drift monitoring** on market_index and rate distributions in production.

18. Conclusion

The delivered model predicts freight rates with \$114.99 MAE and 5.03% MAPE on strictly out-of-sample data, a 90.0% improvement over a constant-rate baseline. The pipeline is reproducible from raw CSVs with a fixed seed, leakage-free by construction, and validated end-to-end against the official scorer.

The work's main strength is that its conclusions are measured rather than assumed: the temporal split reflects the real prediction task, the smearing correction was tested before adoption, and the two findings most likely to be misread — the flat December curve and the weak market_index effect — were investigated and explained rather than presented at face value.